

CSE499A – Section 15 – Group 03

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CityPulse: Proposed AI Methodology

*Parking Slot Occupancy Detection — MobileNetV2
Fine-Tuning Approach*

I. OVERVIEW

CityPulse detects whether each parking slot is occupied or empty using images cropped from existing CCTV cameras installed at registered parking spaces. This is a binary classification task: one patch per slot, one label per patch — *occupied* or *empty*.

The baseline reference for this work is Prova et al. (IEEE IEMTRONICS 2022) [1], who proposed a custom binary CNN trained on a small subset of PKLot and tested directly on Bangladeshi parking footage. That model achieved 86–90% accuracy on local footage with no local training. This work improves on that baseline in three dimensions: model architecture (pretrained transfer learning instead of a custom CNN from scratch), dataset scale (808k images versus 69k), and split strategy (lot-based generalization instead of day-based), before any local fine-tuning is applied.

II. COMPARISON WITH PRIOR WORK

TABLE I
PROPOSED APPROACH VS. PROVA ET AL. (2022)

Aspect	Prova et al.	This Work
Model	Custom binary CNN (from scratch)	MobileNetV2 (ImageNet pretrained, fine-tuned)
Training data	PKLot 2-day subset (~69k)	Full PKLot + CNRPark+EXT (808k)
PKLot split	Day-based (2 training days)	Lot-based (PUCPR + UFPR04 train, UFPR05 val)
CNRPark data	~12k patches	113k patches (official splits)
Pretrained weights	None	ImageNet (1.28M images)
Local fine-tuning	Not explored	Planned (data collection pending)
Edge target	Raspberry Pi 4	Raspberry Pi 4 / Pi 5

III. MODEL ARCHITECTURE

A. Why MobileNetV2

A custom CNN trained from scratch must learn low-level visual features entirely from the training data. MobileNetV2 [2] arrives with those features already learned from 1.28 million ImageNet images. The parking occupancy task only requires the network to learn the distinction between an occupied and empty slot on top of a backbone that already understands edges, textures, shapes, and colour gradients.

Empirical results from the parking occupancy literature confirm this. A custom CNN on PKLot and CNRPark achieves around 93% (*mAlexNet* [4]); an improved MobileNetV3 on the same benchmarks achieves 98.01% [6]. The gap is explained by pretraining, not architectural complexity. MobileNetV2 uses depthwise separable convolution blocks, giving $8\text{--}9\times$ fewer multiply-accumulate operations than a standard convolution of equivalent capacity, making real-time inference on Raspberry Pi 4 and Pi 5 feasible.

B. Architecture

The base MobileNetV2 feature extractor produces a 1280-dimensional representation for any 224×224 RGB input. The original 1000-class classifier is replaced with a two-class head:

```
input MobileNetV2.features (frozen in Phase 1)
  ↓ AdaptiveAvgPool2d
  ↓ Dropout(p=0.2)
  ↓ Linear(1280 → 2)
  ↓ Softmax → [ $P_{\text{empty}}$ ,  $P_{\text{occupied}}$ ]
```

- **MobileNetV2.features** — the pretrained backbone. Frozen in Phase 1 so pretrained ImageNet weights are not disturbed while the new head is initializing.
- **AdaptiveAvgPool2d** — collapses the spatial feature map to a fixed 1280-d vector regardless of input size.
- **Dropout(p=0.2)** — zeroes 20% of activations during training to reduce overfitting on the parking domain.
- **Linear(1280 → 2)** — the new classifier head, trained from scratch in Phase 1. Maps features to two logits.
- **Softmax** — converts logits to probabilities summing to 1. The `argmax` gives the predicted label; the winning probability is stored as the confidence score per slot.

IV. DATASET

The training dataset is `raahad/parking-occupancy-merged` [5], combining PKLot [3] and CNRPark+EXT [4] into a single ImageFolder-compatible directory tree.

TABLE II
DATASET STATISTICS

Split	Empty	Occupied	Total
train	336,589	287,970	624,559
val	73,591	110,841	184,432
Total	410,180	398,811	808,991

PKLot (695,851 patches) — three lots in Brazil (PUCPR, UFPR04, UFPR05) across Sunny, Overcast, and Rainy conditions. Images captured every 5 minutes; same-day frames are near-identical.

CNRPark+EXT (113,140 patches) — nine camera angles in Parma, Italy, across three weather conditions, using the official `LABELS/train.txt` and `val.txt` splits.

A. Split Strategy

PKLot uses a **lot-based split**: PUCPR and UFPR04 train; UFPR05 validates. This forces generalization to an unseen parking lot and camera angle — a stricter test than the day-based split used by Prova et al., which risks data leakage from near-duplicate frames taken 5 minutes apart on the same recording day.

V. TRAINING PROCEDURE

Training follows a two-phase strategy to preserve pretrained weights while adapting the network to the parking occupancy domain.

Phase 1 — Head Training. The MobileNetV2 backbone is frozen. Only the new `Linear(1280→2)` classifier is trained. This gets the head to a reasonable starting point without disturbing the pretrained features. Hyperparameters: Adam optimizer, learning rate 10^{-3} , batch size 128, 5 epochs, single GPU (Tesla T4 on Kaggle), `num_workers=0` to prevent RAM growth under memory-constrained training environments.

TABLE III
PHASE 1 RESULTS — VAL ON UFPR05 (UNSEEN LOT)

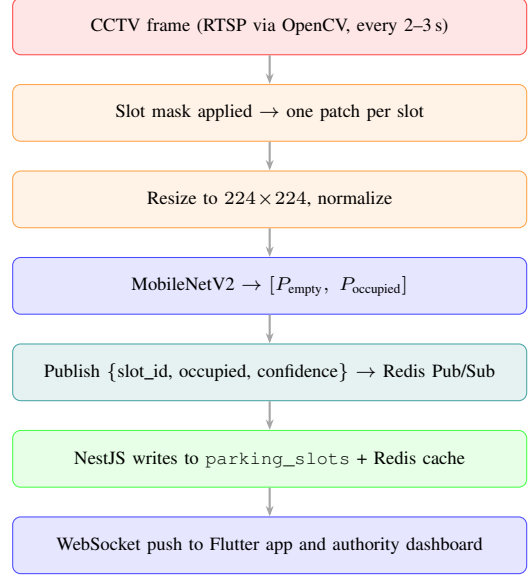
Epoch	Tr. Loss	Tr. Acc	Val Loss	Val Acc
1	0.0466	0.9848	0.1094	0.9631
2	0.0421	0.9862	0.0916	0.9698
3	0.0423	0.9861	0.1057	0.9655
4	0.0414	0.9864	0.1178	0.9624
5	0.0416	0.9866	0.0803	0.9728

Best checkpoint: epoch 5, val accuracy **0.9728**. The validation accuracy does not degrade monotonically between epochs because UFPR05 is from a different lot than the training data — the model generalizes but the gap reflects distribution shift, not overfitting.

Phase 2 — Full Fine-Tuning. All layers are unfrozen and trained end-to-end at a lower learning rate (10^{-4}), allowing the backbone to adapt to parking-specific visual patterns while preserving the pretrained initialization. Runs for 10 epochs; best checkpoint saved by validation accuracy. Phase 2 is pending local data collection.

Input normalization uses ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Augmentation is light — random horizontal flip and mild colour jitter — since the dataset already covers diverse weather, lighting, and camera angles across two source datasets spanning two continents.

VI. INFERENCE PIPELINE



Slot masks are defined once per camera by the parking space owner via the registration portal and stored as JSONB polygons in the `parking_slots.mask_polygon` column. The AI service loads these coordinates on camera startup and uses them to crop per-slot patches from each RTSP frame. This manual mask step is unavoidable — there is no general model that auto-segments slot boundaries from an arbitrary camera position [1]. The FastAPI inference service (`POST /infer`) is operational and accepts single-slot images returning label and confidence. The RTSP ingestion worker and Redis event pipeline are planned for Sprint 4.

VII. RESULTS AND EXPECTED TARGETS

Phase 1 training is complete. The model achieves **97.28%** validation accuracy on UFPR05 after five epochs of head-only training — already surpassing the 93.15% mAlexNet baseline and approaching the 98.01% state-of-the-art. This is achieved before any local Bangladeshi data is included. Phase 2 full fine-tuning is expected to push this higher.

TABLE IV
ACCURACY BENCHMARKS ON PKLOT + CNRPARK+EXT

Model	Acc.	Note
mAlexNet [4]	93.15%	Custom CNN baseline
Prova et al. [1]	86–90%	Dhaka footage, no local training
MobileNetV3 improved [6]	98.01%	Best published result
This work — Phase 1	97.28%	Head-only, benchmark data
This work — Phase 2	≥97.5%	Full fine-tune (pending)

VIII. INTEGRATION WITH CITYPULSE

The AI inference runs as a dedicated FastAPI microservice. It communicates with the NestJS backend via two channels:

Redis Pub/Sub for asynchronous slot status events after each inference cycle, and **internal HTTP** for synchronous calls such as camera registration at owner onboarding. Inference results are also logged to MongoDB with slot ID, space ID, confidence score, and frame timestamp. A 30-day TTL index on the detection events collection keeps storage bounded.

The `parking_slots.occupied` flag and `confidence_score` column are updated in PostgreSQL after each inference cycle so that the booking system can read real-time slot availability without querying MongoDB.

IX. LOCAL FINE-TUNING (PLANNED)

Prova et al. achieved 86–90% on Dhaka footage using a model trained only on international data, with no local fine-tuning attempted. Even without local data, Phase 1 of this work already exceeds that baseline on the international benchmark. After collecting and annotating 200–500 local Bangladeshi parking lot images (planned for Sprint 4), Phase 2 will be repeated on a combined dataset. Given the findings in the literature that even small local additions to a pretrained model produce measurable accuracy gains on local footage, this step is expected to close the remaining distribution gap for Dhaka conditions.

REFERENCES

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